

Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d +Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

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We present results on the near-side of high- p_T triggered di-hadron correlations of neutral strange baryons (Λ , $\bar{\Lambda}$) and mesons (K_S^0) at intermediate $p_T = 2$ -6 GeV/ c in d +Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The near-side di-hadron correlation in $(\Delta\phi, \Delta\eta)$ contains two structures, a peak which is narrow in $\Delta\phi$ and $\Delta\eta$ consistent with correlations due to jet fragmentation and correlation in $\Delta\phi$ which is broad in $\Delta\eta$ which is attributed dominantly to flow. The dependence of the conditional yield of the jet-like correlation on the trigger particle momentum, associated particle momentum, and centrality are presented. In addition the particle composition of the jet-like correlation is determined using identified associated particles. Data on identified triggers are qualitatively described by PYTHIA, however the composition of the jet-like correlation is not quantitatively described by PYTHIA.

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I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Measurements of nuclear modification factors [1, 2, 3, 4] have shown that the nuclear medium created is opaque to particles with large transverse momentum (p_T). Moreover, anisotropic elliptic flow measurements demonstrate that the medium exhibits partonic degrees of freedom and has properties close to those expected of a perfect fluid [5, 6].

Studies of jets in heavy ion collisions are possible through single particle measurements [1, 2, 3, 4], di-hadron correlations [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17], and measurements of reconstructed jets [3, 18, 19, 20, 21]. Measurements of reconstructed jets are useful for studying higher momenta and have provided direct evidence for partonic energy loss in the medium. However, di-hadron correlations provide complementary measurements since they enable studies at intermediate momenta, where the interplay between jets and the medium is important and direct jet reconstruction is no longer possible.

Properties of jets have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum at the Relativistic Heavy Ion Collider (RHIC) [7, 8, 9, 10, 11, 12, 13, 14] and the Large Hadron Collider (LHC) [15, 16, 17]. Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed significant modification, including the disappearance of the peak at intermediate p_T (> 2 GeV/ c) [10, 22] and its reappearance at high p_T [11, 23]. The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central Au+Au collisions [8, 12, 24]. In p + p and d +Au collisions, there is a peak that is narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to

as the jet-like correlation. In Cu+Cu and Au+Au collisions this peak is observed to be broader than that in d +Au collisions, although the yields are comparable [7]. This indicates that the jet-like correlation is dominantly produced through fragmentation.

Besides the shape modifications of jet-like correlations at intermediate transverse momenta, the production mechanism of hadrons may differ from simple fragmentation. In central Au+Au collisions baryon production in the bulk is enhanced relative to that in p + p or d +Au collisions [25, 26]. The baryon to meson ratios measured in Au+Au collisions increase with p_T until reaching a maximum of ≈ 3 relative to p + p collisions at $p_T \approx 3$ GeV/ c in both the strange and non-strange quark sectors. A fall-off of the baryon to meson ratio is observed for $p_T > 3$ GeV/ c and both the strange and non-strange baryon to meson ratios in Au+Au collisions approach the values measured in p + p collisions at $p_T \approx 6$ GeV/ c .

In this paper, studies of two-particle correlations on the near-side in d +Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment are presented. Results from two-particle correlations in pseudo-rapidity and azimuth for neutral strange baryons (Λ , $\bar{\Lambda}$) and mesons (K_S^0) at intermediate p_T ($2 < p_T < 6$ GeV/ c) in the different collision systems are compared to unidentified charged hadron correlations (**h-h**). Both identified trigger particles associated with **unidentified** charged particles (K_S^0 -h, Λ -h) and **unidentified** charged trigger particles associated with identified strange particles (h- K_S^0 , h- Λ) are studied. The near-side jet-like yield is studied as a function of centrality of the collision and transverse momentum of trigger and associated particles to look for possible flavor and baryon/meson dependence. The composition of the jet-like correlation is studied using identified associated particles to investigate possible production mechanisms. The results are compared to expectations from PYTHIA.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

STAR [27] is a multi-purpose spectrometer with a full azimuthal coverage consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. This analysis is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [28] with a pseudo-rapidity acceptance $|\eta| < 1.5$. The TPC has in total 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements for each charged particle track. The TPC momentum resolution is determined to be $\Delta p/p \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [28]. Charged particle tracks used in this analysis were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm and a pseudorapidity $|\eta| < 1.0$. The results presented in this paper are based on analysis of data from d +Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV taken by the STAR experiment in 2003, 2005, and 2004, respectively.

For d +Au collisions, the events analyzed were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located 18 m from the nominal interaction point in the Au beam direction and accepting $95 \pm 3\%$ of the hadronic cross section [29]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) placed at forward pseudorapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the two ZDCs. The MB Au+Au events required a coincidence between the two ZDCs, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats aligned parallel to the beam axis and arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions. An additional online trigger for central Au+Au collisions was used to sample the most central 12% of the total hadronic cross section. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB.

In order to achieve a more uniform detector acceptance in Cu+Cu and Au+Au data sets, only those events with the primary vertex position along the longitudinal beam direction (z) within 30 cm of the center of the STAR detector were used for the analysis. For the d +Au collisions this vertex position selection was extended to $|z| < 50$ cm. The number of events after the vertex cuts in individual data samples is summarized in Table I.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	No. of events [10^6]
d +Au	0-95%	3
Cu+Cu	0-60%	38
Au+Au	0-80%	28
Au+Au	0-12%	17

We identify weakly decaying neutral strange (V^0) particles Λ , $\bar{\Lambda}$ and K_S^0 by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC as described in [30]:

$$\begin{aligned}\Lambda &\rightarrow p + \pi^-, BR = (63.9 \pm 0.5)\% \\ \bar{\Lambda} &\rightarrow \bar{p} + \pi^+, BR = (63.9 \pm 0.5)\% \\ K_S^0 &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\%\end{aligned}\quad (1)$$

where BR denotes the branching ratio. The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates. Reconstructed Λ and K_S^0 particles are required to be within $|\eta| < 1.0$. Topological cuts are optimized for each data set and chosen to reach a signal to background ratio of at least 15:1. For the analyses presented here, no difference was observed between results with Λ and $\bar{\Lambda}$ trigger particles. Therefore the correlations were added to increase the statistical significance of the results. In the remainder of the discussion the combined particles are referred to simply as Λ baryons.

III. METHOD

A. Correlation technique

The analysis in this paper follows the method in [7]. A high- p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger particle in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) is formed. This distribution, $d^2 N_{\text{raw}}/d\Delta\phi d\Delta\eta$, is normalized by the number of trigger particles, N_{trigger} , and corrected for the efficiency and acceptance of associated tracks:

$$\frac{d^2 N}{d\Delta\phi d\Delta\eta}(\Delta\phi, \Delta\eta) = \frac{1}{N_{\text{trigger}}} \frac{d^2 N_{\text{raw}}}{d\Delta\phi d\Delta\eta} \frac{1}{\varepsilon_{\text{assoc}}(\phi, \eta)} \frac{1}{\varepsilon_{\text{pair}}(\Delta\phi, \Delta\eta)}. \quad (2)$$

The efficiency correction $\varepsilon_{\text{assoc}}(\phi, \eta)$ is a correction for the single particle reconstruction efficiency in TPC and $\varepsilon_{\text{pair}}(\Delta\phi, \Delta\eta)$ is a correction for the finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$. Since the correlations are normalized by the number of trigger particles, the efficiency correction is only applied for the associated particle. The fully corrected correlation functions are averaged between positive and negative $\Delta\phi$ and $\Delta\eta$ regions and are reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$ in the plots.

B. Single charged track efficiency correction

For unidentified charged associated particles, the efficiency correction $\varepsilon_{\text{assoc}}(\phi, \eta)$ is the correction for charged hadrons, identical to that applied in [7]. This single charged track reconstruction efficiency is determined as a function of p_T , η , and centrality by simulating the TPC

response to a particle and embedding the simulated sig-
nals into a real event. The single charged track efficiency
is found to be approximately constant for $p_T > 2$ GeV/ c
and ranges from around 75% for central Au+Au events to
around 85% for peripheral Cu+Cu events. The efficiency
for reconstructing a track in d +Au events is 89%.

For identified associated strange particles, the recon-
struction efficiency $\varepsilon_{\text{assoc}}(\phi, \eta)$ is determined in a similar
way, but forcing the simulated particle to decay through
the channel in Equation 2 and then correcting for the
respective branching ratio. The efficiency for Λ , $\bar{\Lambda}$, and
 K_S^0 ranges from 8% to 15%, increasing with momentum
and decreasing with system size [31]. No correction for
the reconstruction efficiency is applied for identified trig-
ger particles because the reconstruction efficiency does
not vary significantly within the p_T^{trigger} bins used in this
analysis.

The systematic uncertainty associated with the effi-
ciency correction for unidentified associated hadrons, 5%,
is strongly correlated across centralities and p_T bins for
each data set but not between data sets. For identified
associated particle ratios the systematic errors on the ef-
ficiency correction partially cancel out and are negligible
compared to the statistical uncertainties.

For the inclusive spectra the feeddown correction due
to secondary Λ baryons from Ξ baryon decays is 15%,
independent of p_T [32]. For identified Λ trigger particles,
we assume that feeddown lambdas do not change the
correlation since little trigger particle type dependence is
observed in the data. This is consistent with the available
data within errors. For identified associated particles,
we assume the same correlation between primary and
secondary Λ baryons and correct the yield of Λ associated
particles by reducing the yield by 15%.

C. Pair acceptance correction

The requirement that each track falls within $|\eta| < 1.0$
in TPC results in a limited acceptance for track pairs.
The geometric acceptance for a track pair is $\approx 100\%$
for $\Delta\eta \approx 0$ and close to 0% near $\Delta\eta \approx 2$. The track
pair acceptance is limited in azimuth by the 12 TPC
sector boundaries, leading to dips in the acceptance of
track pairs in $\Delta\phi$. To correct for the limited geometric
acceptance, a mixed event analysis was performed using
trigger particles from one event combined with associated
particles from another event, as done in [12]. The event
vertices were required to be within 2 cm of each other
along the beam axis and the events were required to have
the same charged particle multiplicity within 10 particles.
To increase statistics of the mixed event sample, each
event with a trigger particle was mixed with ten other
events.

D. Track merging correction

The track merging effect in unidentified hadron (h)
correlations discussed in [7] is also present for V^0 -h and
h- V^0 correlations. This effect leads to a dip at small $\Delta\phi$
and $\Delta\eta$ due to overlap between the trigger and associ-
ated particles. When one of the particles is a V^0 , this
overlap is between one of the V^0 daughter particles and
the unidentified hadron. The size of the dip due to track
merging depends strongly on the relative momenta of the
particle pair. This effect is larger when the overlapping
particles' momenta are closer. For V^0 -h correlations, the
typical associated particle momentum is approximately
1.5 GeV/ c . Since the K_S^0 decay is symmetric, this means
that the track merging effect is greatest for K_S^0 -h cor-
relations with a trigger K_S^0 momentum of approximately 3
GeV/ c . In a Λ decay, the proton daughter carries more
of the Λ momentum than the pion daughter. Therefore
this effect is larger for Λ trigger particles with lower mo-
menta. Because track merging affects both signal and
background particles and the signal sits on top of a large
combinatorial background, the effect is larger for colli-
sions with a higher charged track multiplicity. Since the
dip in V^0 -h and h- V^0 correlations is the result of a V^0
daughter merged with an unidentified hadron, the dip is
wider in $\Delta\phi$ and $\Delta\eta$ than in unidentified hadron cor-
relations. When the track merging dip is present, it is cor-
rected by fitting a Gaussian to the peak, excluding the
region impacted by track merging, and using the Gaus-
sian fit to extract the yield.

In the absence of a clear dip from track merging, the
yield calculated from bin counting is consistent with the
yield from the Gaussian fit. **The statistical error on the
yield from bin counting is added in quadrature to the sta-
tistical error on the background level from the fit.** When
the residual track merging effect is corrected by a fit,
the track merging correction applied by the fit is approx-
imately the same order as the statistical error on the
yield. We therefore conclude that when no dip is evi-
dent, the track merging effect is negligible compared to
the statistical error on the yield.

E. Yield extraction

A sample 2D correlation function after the corrections
described above is shown in Figure 1. The notation and
method used to extract the yield in this paper follow [7,
12]. The jet-like correlation is narrow in both $\Delta\phi$ and $\Delta\eta$
and is contained within the kinematic cuts in p_T^{trigger} and
 $p_T^{\text{associated}}$ used in this analysis. The di-hadron correlation
from Equation 2 is projected onto the $\Delta\eta$ axis:

$$\left. \frac{dN}{d\Delta\eta} \right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}. \quad (3)$$

All other correlations, including those from v_2 , v_3 , and
higher order flow harmonics, are assumed to be indepen-

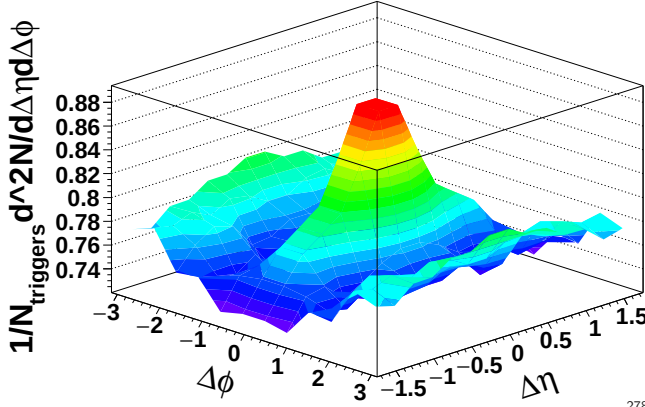


FIG. 1: Corrected 2D K_S^0 -h correlation function for $3 < p_T^{\text{trigger}} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\text{associated}} < p_T^{\text{trigger}}$ for 0-20% Cu+Cu. The data have been reflected about $\Delta\eta = 0$ and $\Delta\phi = 0$.

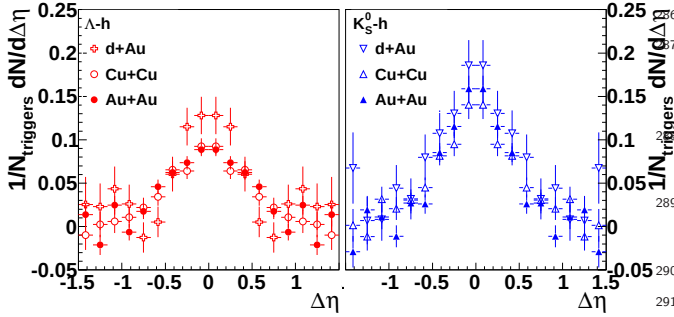


FIG. 2: Corrected correlation functions $\frac{dN_J}{d\Delta\eta}$ in $|\Delta\phi| = 1.0$ for $3 < p_T^{\text{trigger}} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\text{associated}} < p_T^{\text{trigger}}$ for Λ -h (left) and K_S^0 -h (right) for minimum bias d+Au, 0-20% Cu+Cu, and 40-80% Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV after background subtraction. The data have been reflected about $\Delta\phi = 0$.

dent of $\Delta\eta$ within the η acceptance of the analysis, consistent with [12, 33, 34, 35]. We make the assumption that the η dependence observed for v_3 measured using the two-particle cumulant method [36] is entirely due to nonflow. This method explicitly subtracts such correlated backgrounds in addition to uncorrelated background. This means that the jet-like yield can be determined by:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left. \frac{dN}{d\Delta\eta} \right|_{-0.78, 0.78} - b_{\Delta\eta} \quad (4)$$

where $b_{\Delta\eta}$ is a constant offset determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. Variations in the method for extracting the constant background, such as fitting a constant at large $\Delta\eta$, lead to differences in the yield smaller than the statistical error due to the background alone. The extraction of the

background level from a fit can lead to a skew for samples with low statistics. A 2% systematic uncertainty is applied to account for this. This uncertainty is uncorrelated with the uncertainty on the efficiency for a total uncertainty of 5.5% on all yields. Sample correlations are given in Figure 2. When the dip due to track merging is negligible, the yield determined from fit is discarded to avoid any assumptions about the shape of the peak and instead we integrate Equation 4 over $\Delta\eta$ using bin counting to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \frac{dN_J}{d\Delta\eta}(\Delta\eta). \quad (5)$$

Where track merging is negligible the yield from the fit and from bin counting are consistent. Therefore for data where the dip due to track merging is significant, the yield from the Gaussian fit is used, excluding the region affected by track merging. Excluding the peak from the fit region leads to a skew in the extracted yield. For points where the yield is extracted using bin counting, a 13% systematic error was therefore added in quadrature to the statistical error on the yield from the fit so that these points can be compared to the other points.

IV. RESULTS

A. Charged- V^0 correlations

Previous studies demonstrated that the jet-like correlation in h-h correlations is independent of collision system [7, 12, 37] and that it is qualitatively described by PYTHIA [7]. This indicates that the jet-like correlation is dominantly produced by fragmentation, even at intermediate momenta ($2 < p_T < 6$ GeV/c) where recombination is expected to contribute substantially to the bulk production of particles. The composition of the jet-like correlation can be studied using correlations with identified associated particles. For the analysis presented here, the statistics of the d+Au data were limited and the Au+Au data set was limited by the presence of residual track merging. Therefore it was only possible to determine the composition of the jet-like correlation in Cu+Cu collisions.

These measurements are compared to inclusive baryon to meson ratios in $p+p$ collisions from STAR [38] and ALICE [39] and simulations of $p+p$ collisions in PYTHIA [40] using the Perugia 2011 [41] tune and Tune A [42] in Figure 3. The ratio in the jet-like correlation in Cu+Cu collisions is consistent with the inclusive particle ratios from $p+p$. This further supports earlier observations that the jet-like correlation in heavy-ion collisions is dominantly produced by the fragmentation process, which also governs the production of particles in $p+p$ collisions at these momenta. It also implies that production of strange particles through recombination is not significant in the jet-like correlation, even in A+A collisions,

where the inclusive spectra show an enhancement of Λ of up to a factor of three relative to the K_S^0 [32, 43].

The experimental particle ratios at $\sqrt{s} = 200$ and 7000 GeV are consistent with each other, however, they are not described well by PYTHIA. PYTHIA is able to match the light quark meson (π and ω) production [44, 45], but generally underestimates production of strange particles, especially strange baryons [38, 39, 44, 45]. Tune A has been tuned to match low momentum h-h correlations [42], while the Perugia 2011 tune has been tuned to match inclusive particle spectra better, including data from the LHC [41]. The most recent MONASH tune [46], which is a variation of Tune A, had some success in capturing the inclusive strange meson yield at the LHC, but the Λ yield is still underestimated by a factor 2. The discrepancy grows with the strange quark content of the baryon. The problem in PYTHIA is two-fold: since the h- V^0 correlations are dominated by gluon and light quark jet fragmentation, PYTHIA is initially underestimating the generation of strange quarks in those jets. In baryon production this effect gets enhanced since the applied mechanism requires the additional formation of a light di-quark or even a strange di-quark for the formation of a multi-strange baryon. The probability of such a combination is significantly suppressed in PYTHIA, whereas the data seem to suggest that such a production mechanism is not necessary to form strange baryons. Therefore the discrepancy of the PYTHIA Tune A to the data in Figure 3 can be attributed exclusively to the problems of describing strange baryon production in PYTHIA. On the other hand, strange particle triggered correlations, such as K_S^0 -h and Λ -h, originate predominantly from the fragmentation of strange quarks. It should be easier for PYTHIA to describe the fragmentation of an initial strange quark than to describe the production of strange particles from non-strange jets. We therefore studied the V^0 -h correlations in more detail.

B. Correlations with identified strange trigger particles

The jet-like yield as a function of p_T^{trigger} is shown in Figure 4 for K_S^0 -h, and Λ -h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data are tabulated in Table II. Due to residual track merging effects, fits are used for Λ -h correlations in Cu+Cu collisions with $2.0 < p_T^{\text{trigger}} < 3.0$ GeV/c, in 0-12% Au+Au collisions for $3.0 < p_T^{\text{trigger}} < 4.5$ GeV/c, and in 40-80% Au+Au collisions for $2.0 < p_T^{\text{trigger}} < 3.0$ GeV/c. There is no significant difference in the yields between the collision systems as observed earlier for h-h correlations in [7]. This includes no significant difference between results from Au+Au collisions in 40-80% and 0-12% central collisions, although there is a hint that the Λ -h jet-like yield may be lower in central Au+Au collisions, particularly at high p_T . For this reason we only compare to h-h correlations from 40-80% Au+Au collisions.

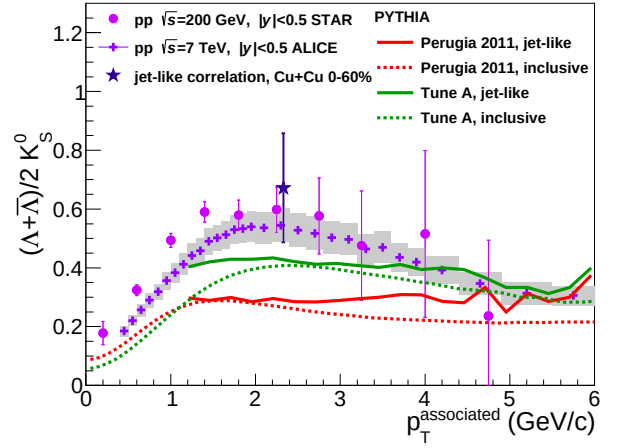


FIG. 3: Λ/K_S^0 ratio measured in the jet-like correlation in 0-60% Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV for $3 < p_T^{\text{trigger}} < 6$ GeV/c and $2.0 < p_T^{\text{associated}} < 3.0$ GeV/c along with this ratio obtained from inclusive p_T spectra in p+p collisions. Data are compared to calculations from PYTHIA [40] using the Perugia 2011 tunes [41] and Tune A [42]. The systematic errors due to the uncertainty on the associated particle's efficiency (5%) and background level extraction (2%) are not shown.

sions.

Next the jet-like yields are studied as a function of collision centrality expressed in terms of number of participating nucleons (N_{part}) calculated from the Glauber model [47]. The extracted jet-like yield as a function of N_{part} is shown in Figure 5 for h-h [7], K_S^0 -h, and Λ -h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. While there is no centrality dependence in the jet-like yield of h-h correlations, there is a centrality dependence in the yields of the K_S^0 -h and Λ -h correlations. These data are compared to PYTHIA [40] calculations from the Perugia 2011 [41] tune in Figure 5. There is a hint of a particle type ordering, with the jet-like yield from K_S^0 -h correlations generally above that of the jet-like yield from h-h correlations and the jet-like yield from Λ -h generally below that of the h-h correlations. This is different from the particle type ordering observed in PYTHIA.

The jet-like yield as a function of $p_T^{\text{associated}}$ is shown in Figure 6 for K_S^0 -h, and Λ -h correlations for d+Au and Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. The Λ -h and K_S^0 -h correlations are only shown for d+Au and Cu+Cu collisions since residual track merging made measurements in Au+Au collisions difficult. Data are compared to the jet-like yield from h-h correlations [7]. The trend is similar for h-h, K_S^0 -h, and Λ -h correlations, however, there are hints that there may be some difference between systems.

TABLE II: The jet-like yield in $|\Delta\eta| < 0.78$ as a function of p_T^{trigger} for K_S^0 -h, and Λ -h correlations for $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$ in minimum bias d +Au, 0-60% Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$, as shown in Figure 4.

p_T	K_S^0 -h				Λ -h			
	MB d +Au	0-60% Cu+Cu	40-80% Au+Au	0-12% Au+Au	MB d +Au	0-60% Cu+Cu	40-80% Au+Au	0-12% Au+Au
2.0-2.5	—	0.036 ± 0.004	0.063 ± 0.008	—	—	0.026 ± 0.005	—	—
2.5-3.0	—	0.059 ± 0.006	0.084 ± 0.023	—	—	0.071 ± 0.007	0.061 ± 0.010	—
3.0-3.5		0.098 ± 0.009	0.139 ± 0.022	—		0.084 ± 0.017	—	—
3.5-4.0	0.162 ± 0.028		0.172 ± 0.021	0.160 ± 0.036	0.079 ± 0.018		0.096 ± 0.030	—
4.0-4.5		0.144 ± 0.011				0.142 ± 0.013		—
4.5-5.0			0.170 ± 0.037	0.240 ± 0.045			0.184 ± 0.040	0.091 ± 0.033
5.0-5.5	—	—			—	—		

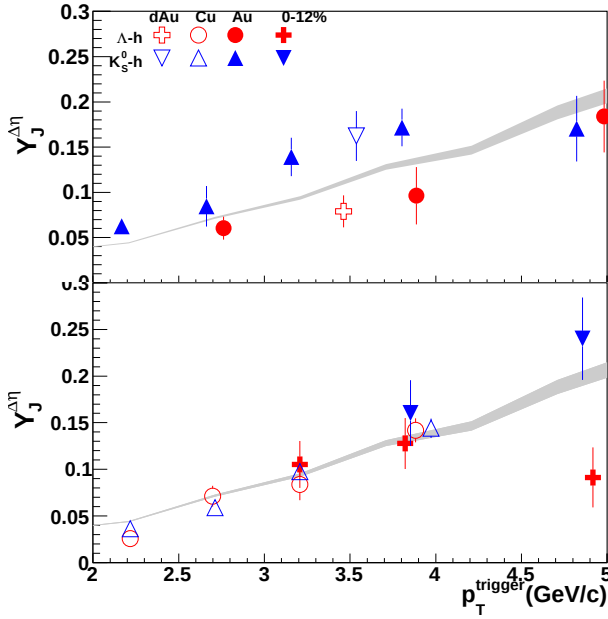


FIG. 4: The jet-like yield in $|\Delta\eta| < 0.78$ as a function of p_T^{trigger} for K_S^0 -h, and Λ -h correlations for $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$ in minimum bias d+Au, 0-60% Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. For comparison h-h correlations [7] from 40-80% Au+Au collisions are shown as a band where the width represents the error. Peripheral Au+Au points have been shifted in p_T^{trigger} for visibility. The systematic errors due to the uncertainty on the associated particle's efficiency (5%) and background level extraction (2%) are not shown.

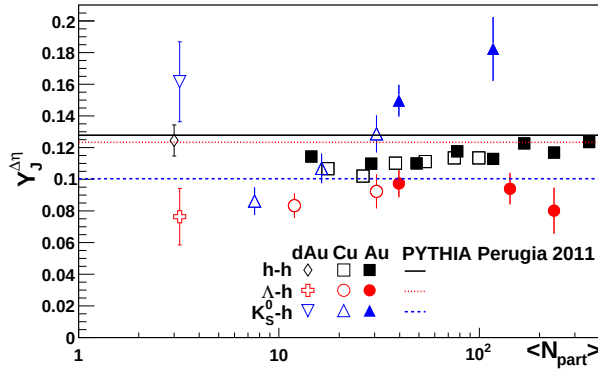


FIG. 5: Centrality dependence of the jet-like yield of K_S^0 -h, and Λ -h correlations for $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$ and $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$ in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$. The data are compared to PYTHIA [40] calculations using the Perugia 2011 tune [41]. The systematic errors due to the uncertainty on the associated particle's efficiency (5%) and background level extraction (2%) are not shown.

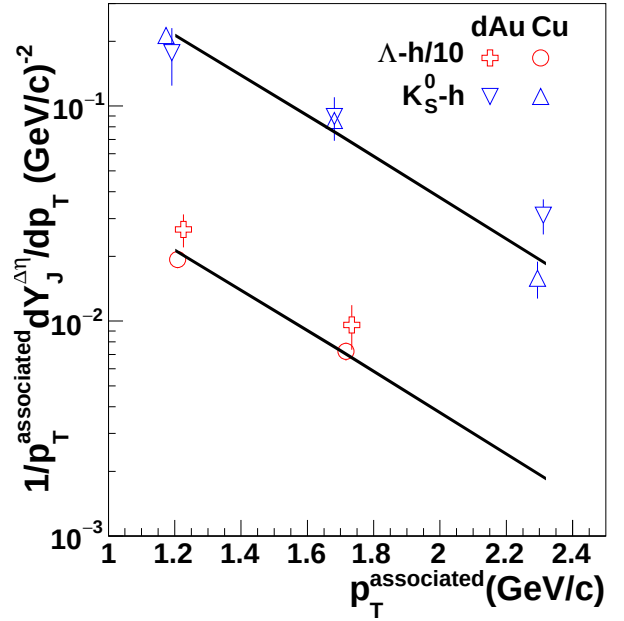


FIG. 6: The jet-like yield as a function of $p_T^{\text{associated}}$ for K_S^0 -h, and Λ -h correlations for $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$ in d+Au and 0-60% Cu+Cu collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$. The data are compared to the jet-like yield from h-h correlations [7] from 40-80% Au+Au collisions are shown as a band where the width represents the error. Data are binned in $1.0 < p_T^{\text{associated}} < 1.5 \text{ GeV}/c$, $1.5 < p_T^{\text{associated}} < 2.0 \text{ GeV}/c$, and $2.0 < p_T^{\text{associated}} < 3.0 \text{ GeV}/c$ and are plotted at the mean of the bin. The systematic errors due to the uncertainty on the associated particle's efficiency (5%) and background level extraction (2%) are not shown.

V. CONCLUSIONS

The $\frac{\Lambda+\bar{\Lambda}}{2K_S^0}$ ratio for the jet-like correlation is comparable to that observed in $p+p$ collisions, providing further evidence that this correlation is dominantly produced by fragmentation. In PYTHIA, this ratio is comparable to the inclusive ratio. We observe a trend for mass ordering of the yield of identified strange trigger particles in d+Au, Cu+Cu, and Au+Au collisions. This mass ordering is not the same ordering observed for the jet-like yield in PYTHIA.

These studies provide motivation for future studies of strangeness production in jets. Larger data sets and data from collisions at higher energies could provide more robust tests of the strangeness production mechanism in $p+p$ collisions. Studies in $p+p$ would be essential in order to search for modifications of strangeness production in jets in heavy ion collisions.

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